

Time Series Forecasting of Electricity Demand in the ISO New England Region

Syeda Aqeel

Executive Summary

This study presents application of time series forecasting to predict monthly electricity demand within the ISO New England Control Area, which encompasses the six New England states: Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, and Vermont. The dataset, obtained from the ISO New England Public Data Portal ([ISO-NE Data Portal](#)), spans from January 2017 through December 2024 and focuses specifically on Non-Peak Time Frame Demand. This metric captures baseline electricity consumption outside designated peak periods and serves as a critical input for wholesale market settlement. Reliable forecasts of Non-PTF Demand are essential for operational planning, enhancing grid reliability, supporting market design, and informing energy policy decisions. Initial data exploration revealed seasonal trends, with demand peaking in both summer and winter months, likely driven by heating and cooling requirements. In response to these patterns, the Box-Jenkins methodology was adopted, culminating in the selection of a Seasonal Autoregressive Integrated Moving Average model with the structure $SARIMA(0,1,1) \times (0,1,1)_{12}$. To benchmark performance, forecasts from a Holt-Winters exponential smoothing model were also developed.

Prior to model fitting, the dataset underwent extensive preprocessing. Data across multiple monthly files were merged to form a unified and continuous time series with a standardized "Datetime" index. The data were confirmed to be free of missing values, and boxplot diagnostics revealed a symmetric distribution without extreme outliers, requiring no further adjustment. Achieving stationarity was critical to model specification. Augmented Dickey-Fuller tests were conducted: seasonal differencing at lag 12 yielded a test statistic of -2.8382 ($p = 0.2321$), failing to reject the null hypothesis of non-stationarity. First-order differencing following seasonal differencing produced a test statistic of -5.7529 ($p = 0.01$), successfully indicating stationarity. Autocorrelation Function and Partial Autocorrelation Function plots informed model specification. The ACF displayed spikes at lags 1 and 12, while the PACF dropped off sharply after lag 1, suggesting non-seasonal and seasonal moving average components of order one. The final SARIMA model is specified as:

$$(1 - B)(1 - B^{12})y_t = (1 - 0.8122B)(1 - B^{12})e_t$$

The non-seasonal MA(1) coefficient was estimated at -0.8122 (SE = 0.0720; 95% CI: [-0.953, -0.671]), while the seasonal MA(1) parameter was -1.0000 (SE = 0.1295; 95% CI: [-1.254, -0.746]), confirming their statistical significance. Model selection criteria included AIC = 1256.54, AICc = 1256.85, and BIC = 1263.80, suggesting strong model fit. Twelve-month forecasts for January 2025 till December 2025 were generated. The SARIMA model successfully reproduced historical seasonal patterns. Residual diagnostics confirmed adequacy: Q-Q plots showed close alignment with the normal distribution, and residuals exhibited white noise properties with no autocorrelation. The Holt-Winters additive model captured general seasonality but exhibited less responsiveness to short-term variations. Its residuals showed larger deviations from normality. Forecast accuracy was evaluated using MAE, RMSE, and MAPE. The SARIMA model achieved a MAE of 306.37 GWh, RMSE of 381.98 GWh, and MAPE of 3.11%, outperforming the Holt-Winters model, which recorded 395.63 GWh, 485.64 GWh, and 4.01%, respectively.

Overall, $SARIMA(0,1,1) \times (0,1,1)_{12}$ proved more effective in capturing the seasonal and stochastic dynamics of electricity demand. These results underscore its value as a robust tool for planners and policymakers seeking to improve grid operations, capacity planning, and policy design. Nonetheless, a limitation of this analysis is its univariate nature, which excludes potential drivers such as temperature or economic indicators.

1. Introduction

1.1 Context and Purpose of the Analysis

This report applies the Box-Jenkins time series methodology to forecast monthly electricity demand within the Independent System Operator New England Control Area, which encompasses the states of Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, and Vermont. The dataset comprises monthly measurements of Non-Peak Time Frame Demand, expressed in gigawatt-hours. The dataset spans from January 2017 to December 2024, providing 96 monthly observations that capture current system dynamics while avoiding issues associated with structurally different historical data. Non-PTF Demand represents the total energy load subject to wholesale market settlement, incorporating demand from non-dispatchable load assets, station service load assets, and unmetered load assets. Accurate forecasting of electricity demand is critical for optimizing operational planning and system reliability. The primary objective of this analysis is to develop a time series model that provide reliable forecasts to guide future consumption planning.

1.2 Data Description

The dataset comprises monthly observations, with each record containing three variables: the calendar year, the calendar month, and the corresponding total energy consumption, measured in gigawatt-hours (GWh). The data were obtained from the publicly accessible ISO New England Data Portal ([ISO-NE Data Portal](#)), which provides official operational and market datasets for the region. Preliminary analysis of the time series indicates the presence of a pronounced annual seasonal pattern, characterized by peaks in energy consumption during the summer and winter months, likely attributable to cooling and heating demands, respectively.

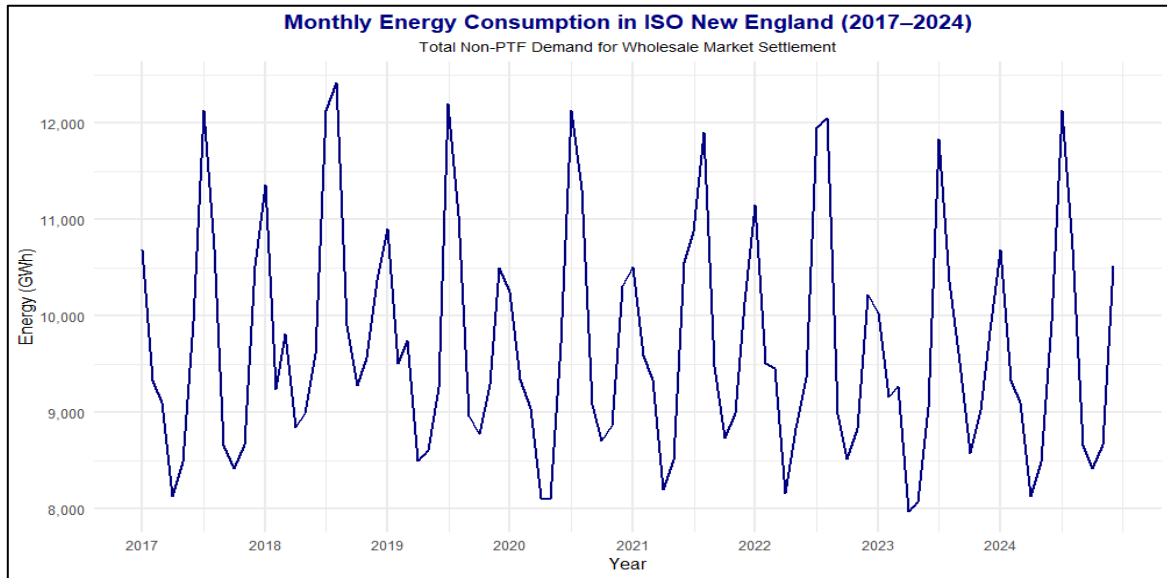


Figure 1: Monthly Energy Consumption in ISO New England (2017 – 2024)

1.3 Data Cleaning and Preprocessing

Prior to model development, the dataset was cleaned to ensure the validity and reliability of the analysis.

1.3.1 File Integration

The energy consumption data spanning January 2017 to December 2024 were stored across multiple files. These files were merged to create a unified dataset suitable for analysis.

1.3.2 Construction of the Datetime Variable

A "Datetime" variable was created by combining the "Year" and "Month" columns, resulting in a continuous monthly index. This transformation was important for structuring the data appropriately for time series modelling.

1.3.3 Missing Values

A review of the monthly observations confirmed that there were no missing entries between January 2017 and December 2024.

1.3.4 Outlier Detection

The dataset was assessed for the presence of outliers. A boxplot of energy consumption values revealed that the data were symmetrically distributed, with no extreme outliers beyond the expected range of variability. The spread between the first and third quartiles was moderate, and the whiskers extended without indicating any anomalous high or low values. Consequently, no adjustments for outliers were necessary.

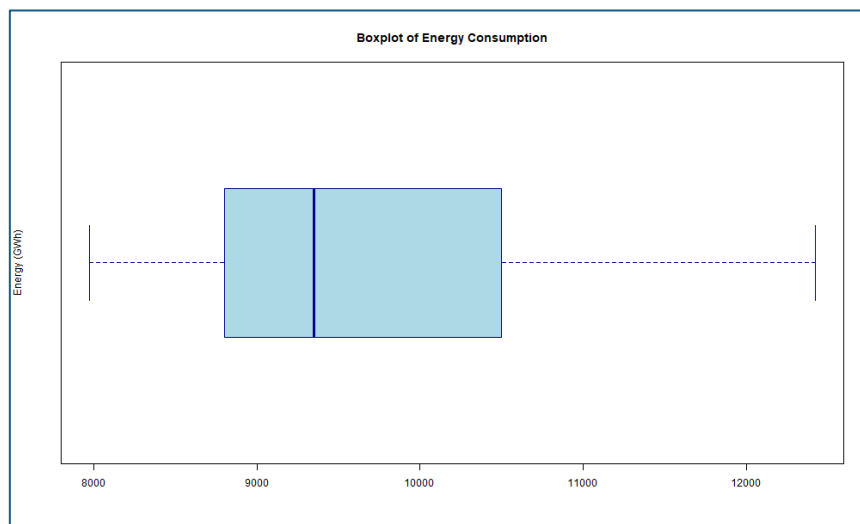


Figure 2: Boxplot of Energy Consumption

2. Model Selection and Justification

The modeling process followed the Box-Jenkins methodology, comprising stationarity testing, model identification, parameter estimation, and model selection. An initial visual inspection of the raw time series suggested the presence of seasonal patterns, indicating non-stationarity. To formally assess stationarity, the Augmented Dickey-Fuller test was conducted on the seasonally differenced and then on the first-order differenced series. The results of the ADF tests are summarized in Table 1. Seasonal differencing alone was insufficient to achieve stationarity, as indicated by a p-value of 0.2321, failing to reject the null hypothesis of a unit root. However, after applying first-order non-seasonal differencing, the ADF test produced a p-value of 0.01, leading to rejection of the null hypothesis and confirming stationarity at the 5% significance level.

Data	Test Statistic	Lag Order	p – value	Decision
Seasonal difference	-2.8382	4	0.2321	Non-Stationary (Fail to Reject Null)
First order non-seasonal difference	-5.7529	4	0.01	Stationary (Reject Null)

Table 1: Augmented Dickey-Fuller test

The plot of the differenced series further corroborates the stationarity of the transformed data. The series fluctuates around a constant mean with no evident trend, and the variance appears relatively stable across time.

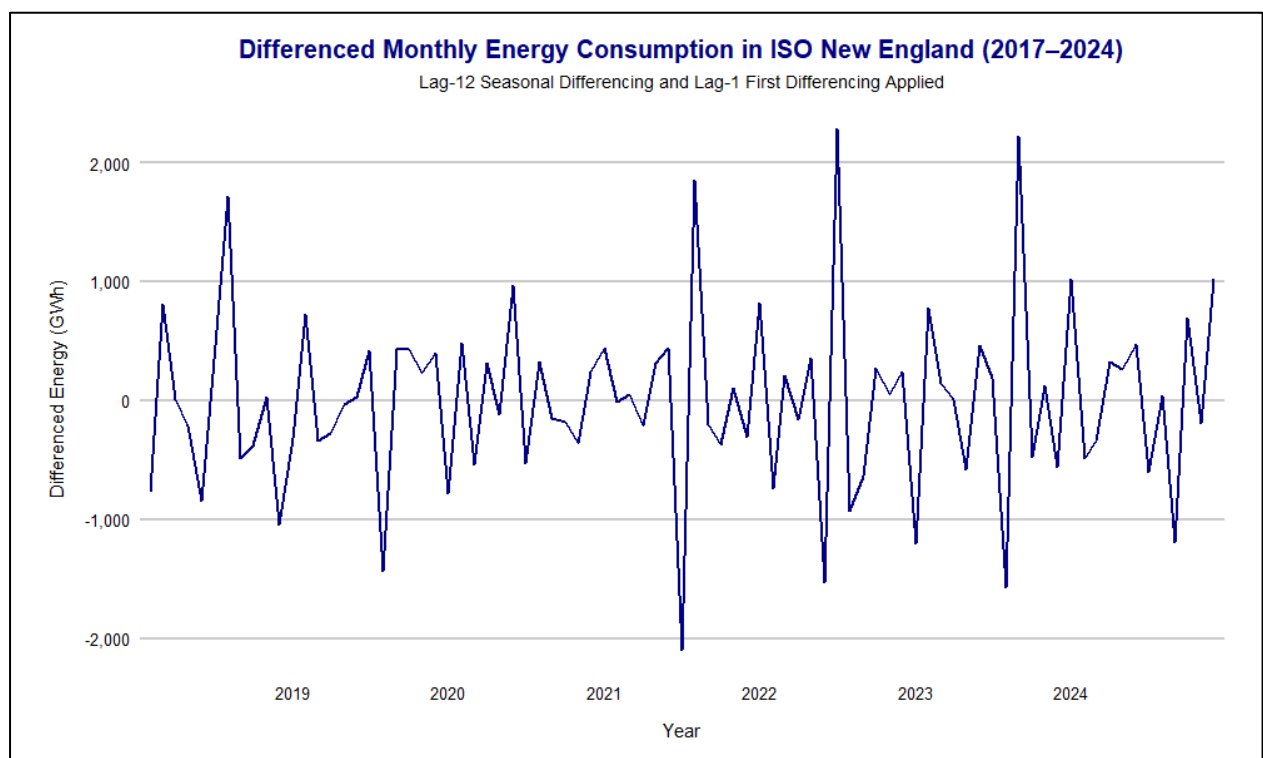


Figure 3: Differenced Monthly Energy Consumption in ISO New England (2017 – 2024)

The autocorrelation function and partial autocorrelation function plots of the differenced series were used to guide the model identification. The ACF exhibited a significant spike at lag 1 with a slow decay, while the PACF cut off after lag 1. Seasonal patterns were evident at multiples of 12 lags, with notable autocorrelations at lag 12. These characteristics suggested a seasonal moving average component of order one and a non-seasonal moving average component of order one

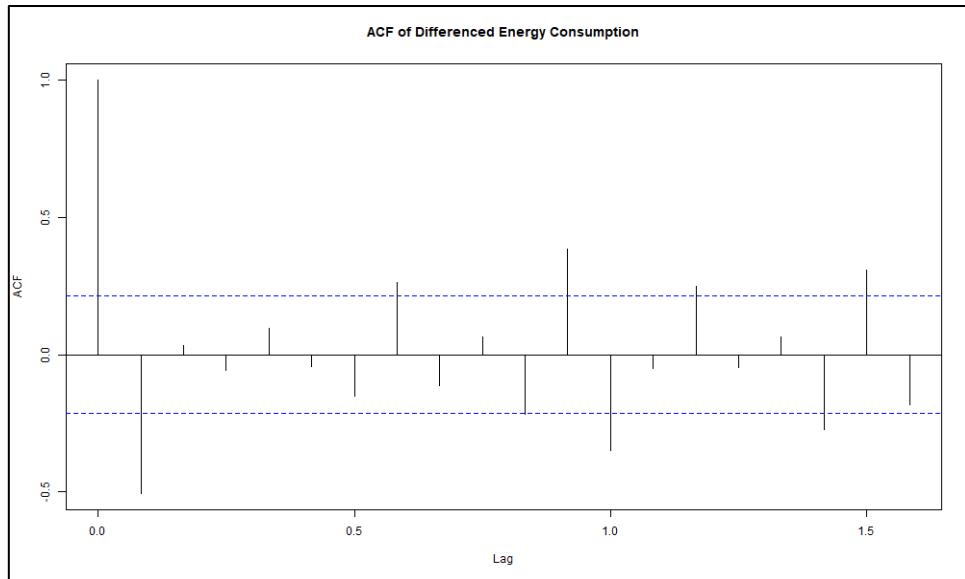


Figure 4: ACF of Differenced Energy Consumption

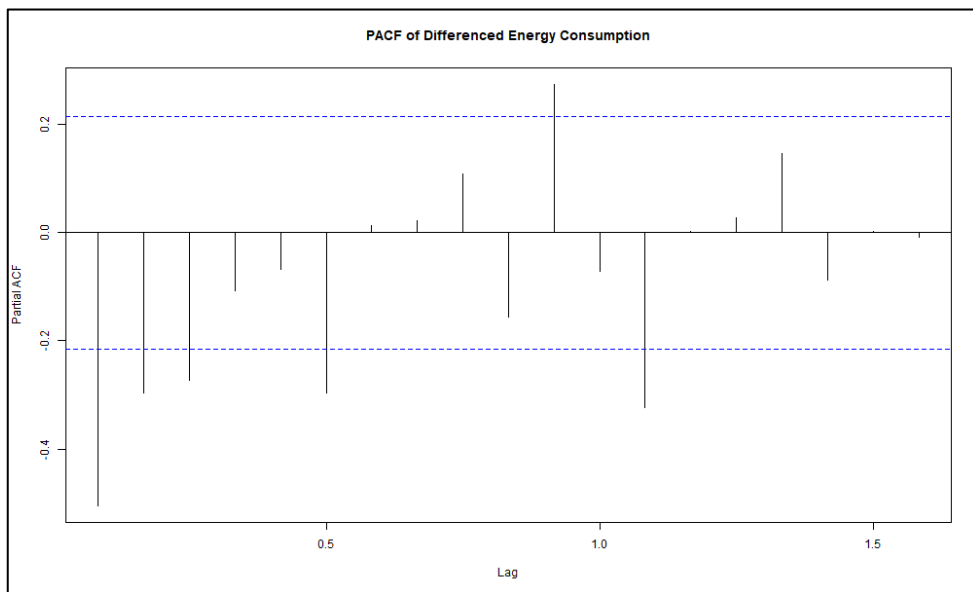


Figure 5: PACF of Differenced Energy Consumption

Based on these diagnostics, the $\text{SARIMA}(0,1,1) \times (0,1,1)_{12}$ model was selected because the underlying characteristics of the data exhibit both non-stationarity and pronounced seasonal patterns. In contrast, a simple ARIMA model without seasonal components would have failed to capture the observed periodic behavior, leading to biased forecasts. The Holt-Winters exponential smoothing method was employed alongside SARIMA modeling to provide a benchmark comparison.

Parameter	Estimate	Standard Error	95% Confidence Interval
Non-seasonal MA(1)	-0.8122	0.0720	[-0.953, -0.671]
Seasonal MA (1)	-1.0000	0.1295	[-1.254, -0.746]

Table 2: Estimated Parameters for SARIMA(0,1,1) × (0,1,1)₁₂ Model

AIC	AICc	BIC
1256.54	1256.85	1263.8

Table 3: Model Selection Criteria for SARIMA(0,1,1) × (0,1,1)₁₂ Model

3. Forecasting and Comparison

3.1 SARIMA Forecasting

$$(1 - B)(1 - B^{12})y_t = (1 - 0.8122B)(1 - B^{12})e_t$$

The SARIMA(0,1,1) × (0,1,1)₁₂ model was utilized to forecast monthly electricity demand for 12 steps ahead, covering the period from January 2025 to December 2025. The forecasts replicate the underlying seasonal patterns present in the historical data, with demand peaks occurring during winter and summer months corresponding to heating and cooling requirements. The forecasted series, along with 80% and 95% prediction intervals, is depicted in Figure 6. The prediction intervals expand moderately over the forecast horizon, reflecting increasing forecast uncertainty over time.

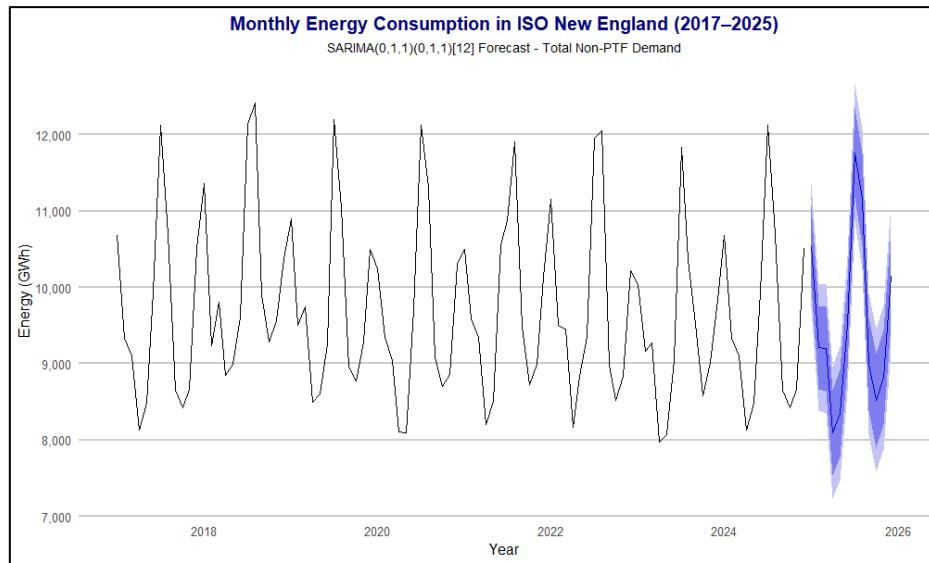


Figure 6: SARIMA(0,1,1) × (0,1,1)₁₂ Forecast

The adequacy of the SARIMA model was further evaluated by examining the residuals. The Q-Q plot of residuals, shown in Figure 7, indicate that residuals largely conform to the theoretical quantiles, supporting the model's validity.

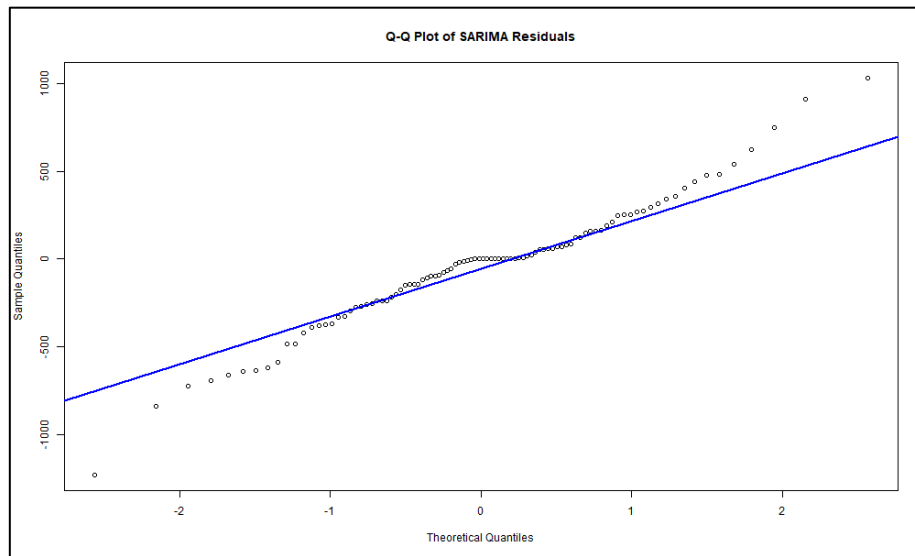


Figure 7: Q-Q Plot of SARIMA Residuals

3.2 Holt Winters Forecasting

As an alternative forecasting method, the Holt-Winters exponential smoothing model with additive seasonality was applied to the same dataset. The Holt-Winters model generated forecasts that similarly captured the seasonal peaks and troughs but exhibited slightly smoother transitions compared to the SARIMA model. The forecasted values and associated prediction intervals are displayed in Figure 8. Compared to SARIMA, the Holt-Winters intervals were slightly wider, reflecting a higher level of forecast uncertainty.

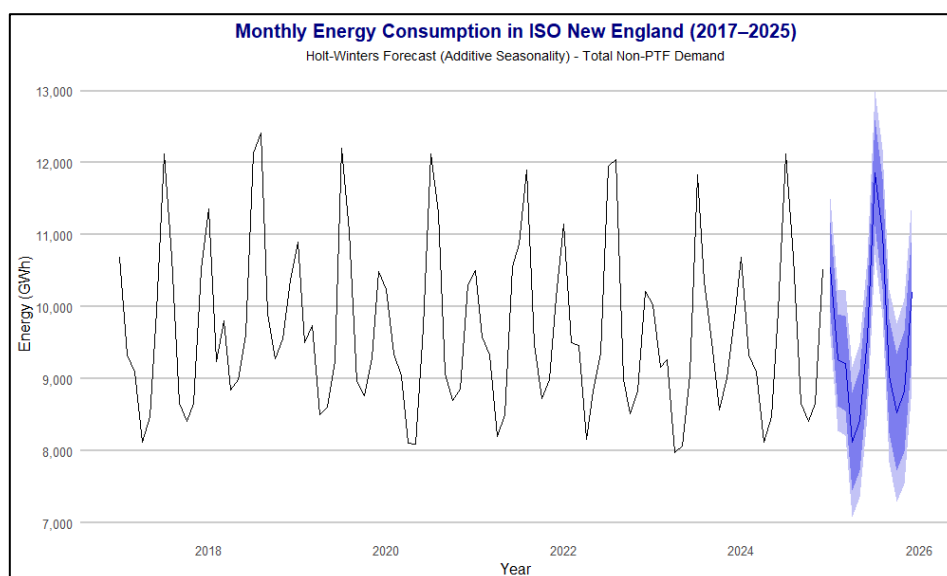


Figure 8: Holt-Winters Forecast

The Q-Q plot for the Holt-Winters residuals, shown in Figure 9, displays greater deviations from the theoretical normal distribution line, particularly in the tails. This suggests that the residuals of the Holt-Winters model are less normally distributed than those of the SARIMA model.

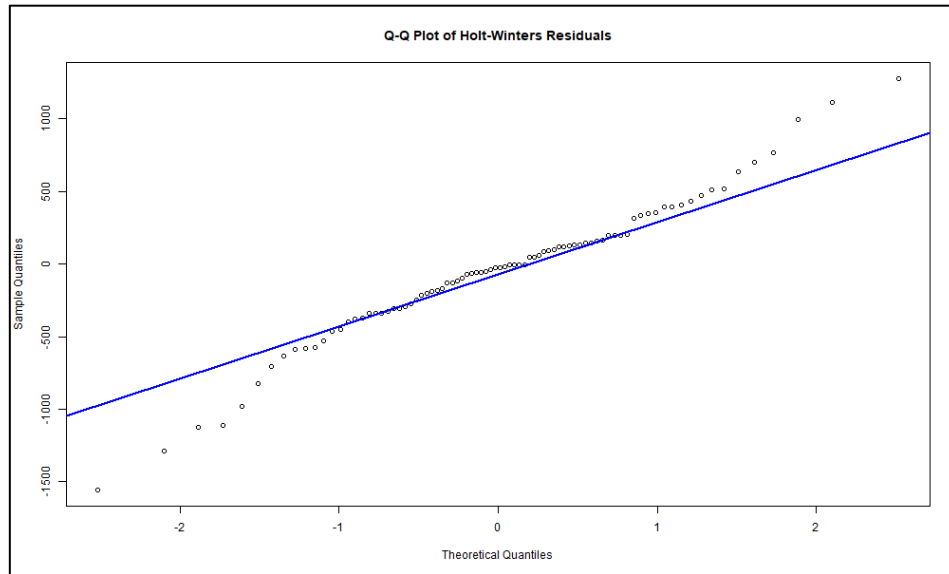


Figure 9: Q-Q Plot of Holt-Winters Residuals

4. Forecast Accuracy Comparison

To evaluate the forecasting ability of the models, a train/test split strategy was employed. The dataset was divided into a training set comprising approximately 80% of the data and a testing set comprising the remaining 20%. Specifically, observations from January 2017 to mid-2023 were used for model training, while subsequent months were reserved for out-of-sample forecast evaluation. Both the $\text{SARIMA}(0,1,1) \times (0,1,1)_{12}$ model and the Holt-Winters additive model were fitted to the training data. Forecasts were generated for the length of the test period and compared to the actual observed values. Forecast performance was assessed using three standard accuracy metrics: Mean Absolute Error, Root Mean Square Error, and Mean Absolute Percentage Error. Forecast plots for both models on the training period are presented in Figures 10 and 11.

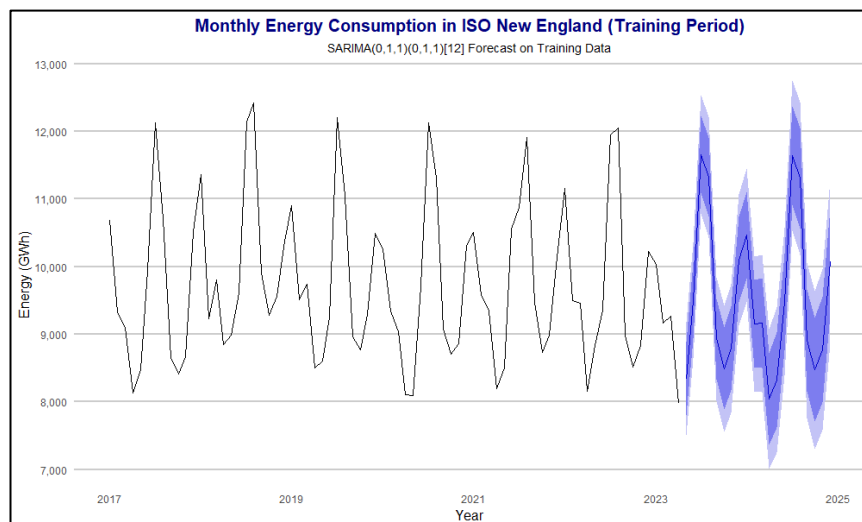


Figure 10: $\text{SARIMA}(0,1,1) \times (0,1,1)_{12}$ Forecast on Training Data

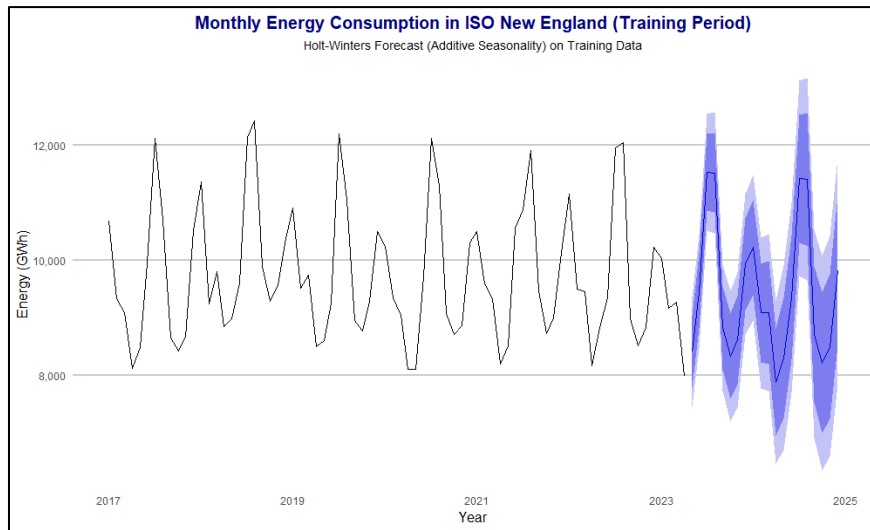


Figure 11: Holt-Winters Forecast on Training Data

The results are summarized in Table 4.

Model	MAE	RMSE	MAPE
SARIMA on Train	306.3660	381.9846	3.112581
Holt-Winters on Train	395.6293	485.6433	4.009925

Table 4: Model Performance Metrics on Training Data

The SARIMA model consistently achieved lower error rates across all three metrics compared to the Holt-Winters model. Specifically, SARIMA reduced the MAE and RMSE by a significant margin and demonstrated a lower mean percentage error. These results reinforce the superior forecasting ability of SARIMA, validating its use as the primary model for predicting electricity demand in the ISO New England region.

5. Findings and Conclusion

This project applied time series analysis to forecast monthly electricity demand for the ISO New England Control Area using a SARIMA(0,1,1) $\times$ (0,1,1)₁₂ model, with Holt-Winters model. Initial exploration of the dataset revealed strong seasonal patterns and non-stationarity, necessitating both seasonal and first-order differencing to achieve stationarity. The SARIMA and Holt Winters model was selected based on its appropriateness for modeling seasonal time series and was fitted using maximum likelihood estimation. Residual analysis confirmed approximate normality and the absence of systematic autocorrelation.

To evaluate model performance, a train/test split was implemented, with approximately 80% of the data used for training and 20% reserved for testing. Forecast accuracy was assessed using Mean Absolute Error, Root Mean Square Error and Mean Absolute Percentage Error. The SARIMA model outperformed the Holt-Winters model across all three metrics, achieving lower forecast errors and

demonstrating superior predictive performance. Specifically, the SARIMA model achieved a MAE of 306.37 GWh, an RMSE of 381.98 GWh, and a MAPE of 3.11%, compared to Holt-Winters' MAE of 395.63 GWh, RMSE of 485.64 GWh, and MAPE of 4.01%. Forecast plots confirmed that the SARIMA model more accurately captured seasonal fluctuations in the data, whereas Holt-Winters forecasts were smoother but less responsive.

Overall, the findings validate the $SARIMA(0,1,1) \times (0,1,1)_{12}$ model as an effective and reliable forecasting tool for ISO New England monthly electricity demand. This model has practical implications for enhancing operational planning, optimizing resource allocation, and supporting market and policy decisions aimed at ensuring grid reliability. However, the current analysis is limited by its univariate structure. Future improvements could extend this analysis by exploring more complex seasonal models, incorporating external regressors such as temperature.

Reference

ISO New England. (2025). *ISO New England public data portal* [Data set]. Retrieved April 29, 2025, from <https://www.iso-ne.com/search?file-type=XLS&file-type=XLSX&file-type=CSV&file-type=xls&file-type=xlsx&file-type=csv&query=data>

Appendix

```
```{r}
library(forecast)
library(tseries)
library(ggplot2)
library(lubridate)
library(scales)

data <- read.csv("C:/Users/shehe/Desktop/ISO.csv")
head(data)
```

```{r}

colnames(data)[colnames(data) == "X...year"] <- "Year"
colnames(data)
```

```{r}
data$Datetime <- as.Date(paste(data$Year, data$Month, "01", sep = "-"))

df <- data.frame(
 Datetime = data$Datetime,
 Energy = data$Energy
)

summary(df)
```

```{r}
colSums(is.na(df))
```
```

```

```{r}
png(filename = "energy_boxplot.png", width = 1000, height = 600)

Q1 <- quantile(df$Energy, 0.25)
Q3 <- quantile(df$Energy, 0.75)
IQR_value <- Q3 - Q1

lower_bound <- Q1 - 1.5 * IQR_value
upper_bound <- Q3 + 1.5 * IQR_value

outliers <- df[df$Energy < lower_bound | df$Energy > upper_bound,]

outliers

boxplot(df$Energy,
 main = "Boxplot of Energy Consumption",
 ylab = "Energy (Gwh)",
 col = "lightblue",
 border = "darkblue",
 horizontal = TRUE)
dev.off()
```

```

```

```{r}
png(filename = "data_plot.png", width = 1000, height = 600)

breaks_year_start <- seq(
 from = floor_date(min(df$Datetime), "year"),
 to = floor_date(max(df$Datetime), "year"),
 by = "1 year"
)

ggplot(df, aes(x = Datetime, y = Energy)) +
 geom_line(size = 1, color = "navyblue") +
 labs(
 title = "Monthly Energy Consumption in ISO New England (2017-2024)",
 subtitle = "Total Non-PTF Demand for Wholesale Market Settlement",
 x = "Year",
 y = "Energy (Gwh)"
) +
 scale_x_date(breaks = breaks_year_start, labels = date_format("%Y")) +
 scale_y_continuous(labels = comma) +
 theme_minimal(base_size = 16) +
 theme(
 plot.title = element_text(face = "bold", size = 20, hjust = 0.5, color = "navyblue"),
 plot.subtitle = element_text(size = 14, hjust = 0.5) # <-- Center the subtitle
)
```

```

```

```{r}
diff_seasonal <- diff(df$Energy, lag = 12)

adf.test(diff_seasonal)
```

```

```

```{r}
diff_final <- diff(diff_seasonal, lag = 1)

adf.test(diff_final)
```

```

```

df_diff <- data.frame(
  Datetime = df$Datetime[-(1:13)],
  Energy_Diff = diff(diff(df$Energy, lag = 12), lag = 1)
)

breaks_year <- seq(
  from = floor_date(min(df_diff$Datetime), "year"),
  to = ceiling_date(max(df_diff$Datetime), "year"),
  by = "1 year"
)

```

```

ggplot(df_diff, aes(x = Datetime, y = Energy_Diff)) +
  geom_line(color = "navyblue", size = 1) +
  labs(
    title = "Differenced Monthly Energy Consumption in ISO New England (2017-2024)",
    subtitle = "Lag-12 Seasonal Differencing and Lag-1 First Differencing Applied",
    x = "Year",
    y = "Differenced Energy (Gwh)"
  ) +
  scale_x_date(
    breaks = breaks_year,
    labels = date_format("%Y"),
    expand = c(0.01, 0.01)
  ) +

```

```

scale_x_date(
  breaks = breaks_year,
  labels = date_format("%Y"),
  expand = c(0.01, 0.01)
) +
scale_y_continuous(labels = comma) + # Clean numeric labels
theme_minimal(base_size = 16) +
theme(
  panel.grid.major.x = element_blank(),
  panel.grid.minor = element_blank(),
  panel.grid.major.y = element_line(color = "gray80"),
  axis.text.x = element_text(size = 13, color = "black"),
  axis.text.y = element_text(size = 13, color = "black"),
  axis.title.x = element_text(size = 15, margin = margin(t = 15)),
  axis.title.y = element_text(size = 15, margin = margin(r = 15)),
  plot.title = element_text(face = "bold", size = 20, hjust = 0.5, color = "navyblue"),
  plot.subtitle = element_text(size = 14, hjust = 0.5, margin = margin(b = 10)),
  plot.margin = margin(20, 30, 20, 30)
)

```

```

{r}
png(filename = "acf_plot.png", width = 1000, height = 600)
energy_ts_diff <- ts(df_diff$Energy_Diff, frequency = 12)
acf(energy_ts_diff, main = "ACF of Differenced Energy Consumption")

```

```

{r}
png(filename = "pacf_plot.png", width = 1000, height = 600)
pacf(energy_ts_diff, main = "PACF of Differenced Energy Consumption")

```

```

{r}
png(filename = "SARIMA_Full_plot.png", width = 1000, height = 600)

energy_ts_full <- ts(
  df$Energy,
  start = c(2017, 1),
  frequency = 12
)

model_sarima_full <- Arima(energy_ts_full, order = c(0,1,1), seasonal = list(order = c(0,1,1), period = 12))
forecast_sarima_full <- forecast(model_sarima_full, h = 12)

```

```

model_sarima_full <- Arima(energy_ts_full, order = c(0,1,1), seasonal = list(order = c(0,1,1), period = 12))
forecast_sarima_full <- forecast(model_sarima_full, h = 12)

autoplot(forecast_sarima_full) +
  labs(
    title = "Monthly Energy Consumption in ISO New England (2017-2025)",
    subtitle = "SARIMA(0,1,1)(0,1,1)[12] Forecast - Total Non-PTF Demand",
    x = "Year",
    y = "Energy (Gwh)"
  ) +
  scale_y_continuous(labels = scales::comma) +
  theme_minimal(base_size = 16) +
  theme(
    plot.title = element_text(face = "bold", size = 20, hjust = 0.5, color = "navyblue"),
    plot.subtitle = element_text(size = 14, hjust = 0.5, margin = margin(b = 10)),
    panel.grid.major.x = element_blank(),
    panel.grid.minor = element_blank(),
    panel.grid.major.y = element_line(color = "gray80")
  )

```

```

{r}
png(filename = "SARIMA_qq.png", width = 1000, height = 600)
residuals_sarima <- residuals(model_sarima_full)

qqnorm(residuals_sarima, main = "Q-Q Plot of SARIMA Residuals")
qqline(residuals_sarima, col = "blue", lwd = 2)

```

```

{r}
model_sarima_full

```

```

{r}
png(filename = "HW_full.png", width = 1000, height = 600)
model_hw_full <- HoltWinters(
  energy_ts_full,
  seasonal = "additive"
)

forecast_hw_full <- forecast(
  model_hw_full,
  h = 12
)

```

```

autoplot(forecast_hw_full) +
  labs(
    title = "Monthly Energy Consumption in ISO New England (2017-2025)",
    subtitle = "Holt-Winters Forecast (Additive Seasonality) - Total Non-PTF Demand",
    x = "Year",
    y = "Energy (GWh)"
  ) +
  scale_y_continuous(
    labels = comma
  ) +
  theme_minimal(base_size = 16) +
  theme(
    plot.title = element_text(
      face = "bold",
      size = 20,
      hjust = 0.5,
      color = "navyblue"
    ),
    plot.subtitle = element_text(
      size = 14,
      hjust = 0.5
    ),
    panel.grid.major.x = element_blank(),
    panel.grid.minor = element_blank(),
    panel.grid.major.y = element_line(color = "gray80"))

```

```

```{r}
png(filename = "hw_qq_plot.png", width = 1000, height = 600)
residuals_hw <- residuals(model_hw_full)

qqnorm(residuals_hw, main = "Q-Q Plot of Holt-Winters Residuals")
qqline(residuals_hw, col = "blue", lwd = 2)
...

```

```

```{r}
n_total <- length(df$Energy)
n_train <- floor(0.8 * n_total)

train_energy <- df$Energy[1:n_train]
test_energy <- df$Energy[(n_train+1):n_total]

train_ts <- ts(train_energy, start = c(2017, 1), frequency = 12)

train_length <- length(train_ts) # in months

start_year_test <- 2017 + (train_length %% 12)
start_month_test <- (train_length %% 12) + 1

test_ts <- ts(
  test_energy,
  start = c(start_year_test, start_month_test),
  frequency = 12
)
...

```

```

```{r}
model_sarima_train <- Arima(train_ts, order = c(0,1,1), seasonal = list(order = c(0,1,1), period = 12))
forecast_sarima_train <- forecast(model_sarima_train, h = length(test_ts))
...

```

```

```{r}
png(filename = "SARIMA_train.png", width = 1000, height = 600)

autoplot(forecast_sarima_train) +
  labs(
    title = "Monthly Energy Consumption in ISO New England (Training Period)",
    subtitle = "SARIMA(0,1,1)(0,1,1)[12] Forecast on Training Data",
    x = "Year",
    y = "Energy (GWh)"
  ) +
  scale_y_continuous(
    labels = comma
  ) +
  theme_minimal(base_size = 16) +
  theme(
    plot.title = element_text(face = "bold", size = 20, hjust = 0.5, color = "navyblue"),
    plot.subtitle = element_text(size = 14, hjust = 0.5),
    axis.text.x = element_text(size = 12),
    axis.text.y = element_text(size = 12),
    panel.grid.major.x = element_blank(),
    panel.grid.minor = element_blank(),
    panel.grid.major.y = element_line(color = "gray80"))

```

```

```{r}
png(filename = "hw_train_plot.png", width = 1000, height = 600)
model_hw_train <- Holtwinters(train_ts, seasonal = "additive")
forecast_hw_train <- forecast(model_hw_train, h = length(test_ts))

```

```

autoplot(forecast_hw_train) +
 labs(
 title = "Monthly Energy Consumption in ISO New England (Training Period)",
 subtitle = "Holt-Winters Forecast (Additive Seasonality) on Training Data",
 x = "Year",
 y = "Energy (GWh)"
) +
 scale_y_continuous(labels = comma) +
 theme_minimal(base_size = 16) +
 theme(
 plot.title = element_text(
 face = "bold",
 size = 20,
 hjust = 0.5,
 color = "navyblue"
),
 plot.subtitle = element_text(
 size = 14,
 hjust = 0.5
),
 axis.text.x = element_text(size = 12),
 axis.text.y = element_text(size = 12),
 panel.grid.major.x = element_blank(),
 panel.grid.minor = element_blank(),
 panel.grid.major.y = element_line(color = "gray80"))

```

```

```{r}

sarima_errors <- as.numeric(test_ts) - forecast_sarima_train$mean
sarima_mae <- mean(abs(sarima_errors))
sarima_rmse <- sqrt(mean((sarima_errors)^2))
sarima_mape <- mean(abs((sarima_errors / as.numeric(test_ts)) * 100))

hw_errors <- as.numeric(test_ts) - forecast_hw_train$mean
hw_mae <- mean(abs(hw_errors))
hw_rmse <- sqrt(mean((hw_errors)^2))
hw_mape <- mean(abs((hw_errors / as.numeric(test_ts)) * 100))

```

```

```

```{r}
data.frame(
  Model = c("SARIMA on Train", "Holt-Winters on Train"),
  MAE = c(sarima_mae, hw_mae),
  RMSE = c(sarima_rmse, hw_rmse),
  MAPE = c(sarima_mape, hw_mape)
)

```